

A stretched tetrahedron disproves the proposed John-position facet-ratio extremum

Candidate counterexample packet for arXiv:2302.01213

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Status. Candidate counterexample, likely valid; human review requested.

1 Source conjecture

The source is Maud Szusterman, *A New Excluding Condition Towards the Soprunov–Zvavitch Conjecture on Bezout-type inequalities*, arXiv:2302.01213 [1]. On page 7, after comparing the regular and standard simplices, the paper proposes that

$$\max_F \frac{\text{Isop}(F)}{\text{Isop}(T_n)} = \min_T \max_F \frac{\text{Isop}(TF)}{\text{Isop}(TT_n)},$$

that is, that the maximal facet/body ratio is minimized when the simplex is in John’s position. Here

$$\text{Isop}(K) = \frac{1}{d} \frac{|\partial K|_{d-1}}{|K|_d} \quad (\dim K = d).$$

The surrounding discussion and examples make clear that the intended minimization is over invertible affine (equivalently, linear after a translation) images. If volume preservation is desired, a uniform rescaling of the map below gives the same ratios.

Nonetheless, one can find an affine transform T such that $\frac{\text{Isop}(TF)}{\text{Isop}(TM)} > 1$, showing that $b_2(M) > 1$. Indeed, let $c > 1$ be large enough and let T be the unique affine transform such that $Te_n = ce_n$, and $Te_i = e_i$ for $i \leq n-1$. Then computations show that $\text{Isop}(TM) = \frac{|\partial \mathcal{E}|}{n|\mathcal{E}|} + \frac{2\kappa_{n-1}}{n\kappa_n c} < \frac{|\partial \mathcal{E}|}{n|\mathcal{E}|} + \frac{1}{c} < 1 = \text{Isop}(TF) = \text{Isop}(F)$, showing $b_2(M) = b_2(TM) < 1$. (see appendix for computational details)

In the above four examples (unit cube, octahedron O_n , cylinders, half-balls), we used Proposition 1 (or Proposition 2) to argue that $b_2(K) > 1$. For these examples, the fact that $b_2(K) > 1$ was already known : the cylinder, like the cube, is decomposable, the half-ball has (many) points of positive curvature on its boundary, and it was directly shown in [SZ15] that $b_2(O_n) = 2$ (by taking well-chosen segments). The half-ball example suggests that $\max_F \frac{\text{Isop}(F)}{\text{Isop}(K)}$ could be minimal when K is in its John's position.

As a sanity check, one may wish to compute $\max_F \frac{\text{Isop}(F)}{\text{Isop}(\Delta)}$, when Δ is an n -simplex, to check this quantity is less than 1. If $\Delta = T_n$ is a regular simplex with edge length $\sqrt{2}$, then $|T_n| = \frac{\sqrt{n+1}}{n!}$, and $|\partial T_n| = (n+1)|T_{n-1}| = (n+1)\frac{\sqrt{n}}{(n-1)!}$, so that $\text{Isop}(T_n) = \sqrt{n(n+1)}$. It results that for a regular n -simplex :

$$\max_F \frac{\text{Isop}(F)}{\text{Isop}(T_n)} = \frac{\sqrt{n(n-1)}}{\sqrt{n(n+1)}} = \sqrt{\frac{n-1}{n+1}}.$$

If $\Delta_n = \text{Conv}(0, e_1, \dots, e_n)$, then $|\Delta_n| = \frac{1}{n!}$, and $|\partial \Delta_n| = \frac{n}{(n-1)!} + \frac{\sqrt{n}}{(n-1)!}$, hence $\text{Isop}(\Delta_n) = n + \sqrt{n}$. There are two kinds of facets : one is Δ_{n-1} , the other one is T_{n-1} . Their isoperimetric ratios are respectively $\text{Isop}(T_{n-1}) = \sqrt{n(n-1)}$ and $\text{Isop}(\Delta_{n-1}) = n-1 + \sqrt{n-1} > \text{Isop}(T_{n-1})$. Therefore :

$$\max_F \frac{\text{Isop}(F)}{\text{Isop}(\Delta_n)} = \frac{\text{Isop}(\Delta_{n-1})}{\text{Isop}(\Delta_n)} = \frac{n-1 + \sqrt{n-1}}{n + \sqrt{n}}.$$

Note that $\max_F \frac{\text{Isop}(F)}{\text{Isop}(\Delta_n)} > \max_F \frac{\text{Isop}(F)}{\text{Isop}(T_n)}$. One may conjecture that $\max_F \frac{\text{Isop}(F)}{\text{Isop}(T_n)} = \min_T \max_F \frac{\text{Isop}(TF)}{\text{Isop}(T\Delta_n)}$, i.e. that the (maximal) ratio is minimal when the simplex is in its John's position.

2 Proof intuition

Every simplex is tangential: its insphere touches every facet. Consequently its isoperimetric quantity is the reciprocal of its inradius. The conjectured facet/body ratio is therefore just the global inradius divided by the inradius of the facet.

Start with a regular tetrahedron and stretch one edge by a factor of 3. There are then only two symmetry classes of facets. Their side lengths, areas, and inradii are elementary. Both resulting ratios are strictly less than the regular value $1/\sqrt{2}$. Thus the stretch makes the proposed objective smaller, rather than larger.

3 Formal counterexample

Lemma 1. *If S is a d -simplex with inradius $r(S)$, then*

$$\text{Isop}(S) = \frac{1}{r(S)}.$$

Hence, for every facet F of an n -simplex S ,

$$\frac{\text{Isop}(F)}{\text{Isop}(S)} = \frac{r(S)}{r(F)}.$$

Proof. Writing $A_0, \dots, A_d$ for the facet areas and V for the volume, decompose S into the $d+1$ pyramids with common apex at the incenter. Then

$$V = \frac{r(S)}{d} \sum_{i=0}^d A_i = \frac{r(S)}{d} |\partial S|_{d-1}.$$

The first identity follows. Apply it once to S and once to F for the second. $\square$

Theorem 1. *Let R be a regular unit-edge tetrahedron and let $T = \text{diag}(3, 1, 1)$, with the first coordinate axis along an edge of R . Then*

$$\max_{F \text{ facet of } R} \frac{\text{Isop}(TF)}{\text{Isop}(TR)} < \frac{1}{\sqrt{2}} = \max_{F \text{ facet of } R} \frac{\text{Isop}(F)}{\text{Isop}(R)}.$$

In particular, the proposed John-position minimization is false already for tetrahedra in dimension 3.

Proof. Use the regular tetrahedron with vertices

$$v_0 = (0, 0, 0), \quad v_1 = (1, 0, 0), \quad v_2 = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \quad v_3 = \left(\frac{1}{2}, \frac{1}{2\sqrt{3}}, \sqrt{\frac{2}{3}}\right).$$

All six mutual distances are 1, and the edge v_0v_1 is on the first coordinate axis. Put $S = TR$. Its volume and surface area are

$$V(S) = \frac{\sqrt{2}}{4}, \quad |\partial S| = \frac{3\sqrt{3} + \sqrt{11}}{2}.$$

Indeed, its four triangular facets occur in the following two pairs:

type	multiplicity	side lengths	area
A	2	$(3, \sqrt{3}, \sqrt{3})$	$3\sqrt{3}/4$
B	2	$(1, \sqrt{3}, \sqrt{3})$	$\sqrt{11}/4$

The displayed areas follow either from a cross product or from Heron's formula. Since $r = 3V/|\partial S|$, the inradius of S is

$$r(S) = \frac{3\sqrt{2}}{2(3\sqrt{3} + \sqrt{11})}.$$

The two facet inradii, using $r_\Delta = A/s$ with s the semiperimeter, are

$$r_A = \frac{3\sqrt{3}}{2(3 + 2\sqrt{3})}, \quad r_B = \frac{\sqrt{11}}{2(1 + 2\sqrt{3})}.$$

Lemma 1 now gives the two possible facet/body ratios:

$$q_A = \frac{r(S)}{r_A} = \sqrt{\frac{2}{3}} \frac{3 + 2\sqrt{3}}{3\sqrt{3} + \sqrt{11}},$$

$$q_B = \frac{r(S)}{r_B} = \frac{3\sqrt{2}(1 + 2\sqrt{3})}{(3\sqrt{3} + \sqrt{11})\sqrt{11}}.$$

Both are strictly below $1/\sqrt{2}$. For the first, the desired inequality is equivalent to

$$2\sqrt{3} + 4 < 3\sqrt{3} + \sqrt{11},$$

which follows from $4 < \sqrt{3} + \sqrt{11}$. For the second it is equivalent to

$$12\sqrt{3} - 5 < 3\sqrt{33}.$$

Both sides are positive, and squaring reduces this to $160 < 120\sqrt{3}$, or $4/3 < \sqrt{3}$.

For a regular tetrahedron, the tetrahedral inradius is $\sqrt{6}/12$ and a facet inradius is $\sqrt{3}/6$, so every ratio equals $1/\sqrt{2}$. Therefore $\max(q_A, q_B) < 1/\sqrt{2}$, proving the claim. $\square$

4 Verification and scope

The exact coordinate, area, volume, inradius, and inequality calculations are independently reproduced by `code/verify.py`. In decimals,

$$q_A \approx 0.6199994121, \quad q_B \approx 0.6708149210, \quad \frac{1}{\sqrt{2}} \approx 0.7071067812.$$

The script uses exact SymPy radicals before the final decimal display.

This disproves only the paper’s explicit simplex/John-position extremal conjecture on page 7. It does not answer Question 1 on page 8 for non-simplex polytopes, and it does not address the broader Soprunov–Zvavitch conjecture. The source sometimes writes $O(n)$ where its examples and prose use general affine transforms. The counterexample addresses that intended affine interpretation; under orthogonal transforms alone the ratio is invariant and the displayed minimization would be tautological.

Cheap run indexes were searched for arXiv:2302.01213 and the core phrases “regular simplex”, “John’s position”, “facet isoperimetric ratio”, and “affine simplex”. A bounded web search on 21 July 2026 used the same exact and near-exact phrases and found the source paper and general reverse isoperimetric references, but no later paper resolving this precise conjecture. This is not an exhaustive novelty certification.

Human-review recommendation. Check the intended domain of the affine minimization in the source, then verify the two facet classes and the conversion $\text{Isop} = 1/r$. These are the only substantive points; the remaining radical inequalities are explicit.

References

- [1] M. Szusterman, *A New Excluding Condition Towards the Soprunov–Zvavitch Conjecture on Bezout-type inequalities*, arXiv:2302.01213, 2023. <https://arxiv.org/abs/2302.01213>